

Problem Set 3

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1 Number theory

Problem 1: Prove/Disprove that $4 \nmid (n^2 + 2)$ for any integer n . Note that “ $\nmid$ ” is notation for “doesn’t divide”.

Problem 2: Prove that 7 divides $2^n + 6 \cdot 9^n$ for any positive number n .

Problem 3: Prove that the product of three consecutive integers is divisible by 6.

Problem 4: Prove that the product of four consecutive integers is divisible by 24.

Problem 5: Use the Chinese Remainder Theorem to solve for x (show and describe all steps):

$$x \cong 1 \pmod{2}$$

$$x \cong 2 \pmod{3}$$

$$x \cong 3 \pmod{5}$$